

SecureEntry OTP

GSM-Based Two-Step Time-Limited Door Access System

SETUP & INITIALIZATION PROCEDURE

This document describes the recommended hardware/software setup and the startup initialization sequence for the SecureEntry project based on the project architecture and workflow. The LPC2148 acts as the central controller and initializes the user interface, external memory, GSM communication, RTC, security functions, and door-control interface before entering the normal LOGIN state.

1. Project Setup

Category	Setup / Requirement
Controller	LPC2148 ARM7 microcontroller (ARM7TDMI-S)
Programming	Embedded C; Keil µVision 4
Display	16×2 LCD connected through GPIO
Input	4×4 keypad for User ID, password, OTP and menu input
External Memory	AT25LC512 / AT25F512A SPI EEPROM through SPI0
Wireless	GSM modem through UART0 at 9600 bps
Time Base	RTC for OTP time association and expiry
Door Control	L293D motor driver + DC motor
Management	External interrupt / management switch
Simulation	Proteus, if simulation is used

2. Interface Configuration

Interface	Device	Important Configuration
GPIO	LCD / Keypad / L293D	Configure required pins as input/output according to firmware.
UART0	GSM modem	P0.0 → TXD0, P0.1 → RXD0; 9600 bps.
SPI0	SPI EEPROM	Configure SPI master communication and EEPROM chip-select.
RTC	RTC module	Initialize time/date before OTP expiry checking.
External Interrupt	Management switch	Configure interrupt source and required trigger mode.

3. Power-ON Initialization Sequence

The following sequence is the recommended logical order from POWER ON to SYSTEM READY. The exact function names may differ in the source code, but the initialization responsibilities remain the same.

1	Power ON LPC2148 starts execution. Reset/startup code executes and program control enters the main application.
2	Initialize LCD Configure the 16×2 LCD interface. Initialize LCD pins and display controller, clear the display, and show the startup/project message.
3	Initialize Keypad Configure the 4×4 keypad. Set keypad rows/columns in the required input/output modes and prepare keypad scanning.
4	Initialize SPI / EEPROM Prepare SPI0 communication with the external EEPROM. Configure SPI master mode, chip-select handling and EEPROM communication routines.
5	Check / Initialize EEPROM Verify that the external memory is accessible. Perform the project's EEPROM diagnostic/initialization procedure and load required stored information.
6	Initialize Management Switch & Interrupt Enable the user-management access mechanism. Configure the external interrupt and management switch so the Admin/User Management mode can be entered.
7	Initialize UART0 Configure serial communication for GSM. Set UART0 for the required communication format; project configuration uses 9600 bps.
8	Initialize GSM Prepare the GSM modem. Check modem communication using the AT interface, configure SMS text mode and configure the required incoming-SMS notification mechanism.
9	Initialize RTC Prepare the real-time clock. Read or set the RTC time so OTP generation and expiry calculations have a valid time reference.
10	Initialize OTP Module Prepare OTP generation and validation. Initialize OTP-related variables/state, expiry timing and the logic used for OTP comparison.
11	Load User Database Load user-related information from EEPROM. Retrieve stored user credentials/status information and other required system data.

System Ready**Enter the normal operating state.**

Display the LOGIN prompt and wait for the user to enter the 4-digit User ID.

4. GSM Setup & Initialization

The GSM modem is connected to UART0. The modem must be responding before the system relies on it for OTP delivery or SMS-based BLOCK / UNBLOCK control.

UART0 connection

P0.0 → TXD0
P0.1 → RXD0
Baud Rate → 9600 bps

Basic GSM initialization sequence

Stage	Purpose
AT	Check whether the modem responds; expected response is OK.
AT+CMGF=1	Select SMS Text Mode.
AT+CNMI=1,2,0,0,0	Configure the modem to provide incoming SMS information through the UART interface as required by the project.

5. EEPROM Initialization & Diagnostic

The external SPI EEPROM stores persistent information such as user credentials, user status, failed-attempt information, admin password and registered mobile-number information. The project also includes a communication diagnostic.

Diagnostic concept

WREN → RDSR → Check WEL Bit

RDSR uses the EEPROM status-register command 0x05. If the EEPROM communication check fails, the project can report an EEPROM error on the LCD so that wiring or communication problems can be investigated before normal access operation.

6. RTC & OTP Initialization

The RTC provides the time reference used to associate an OTP with a generation time and to determine whether the OTP has expired. After the password is verified, a 6-digit OTP is generated, sent through GSM, and later validated together with the RTC expiry condition.

OTP operating sequence

Password Correct → Generate 6-Digit OTP → Record/Start OTP Timing → Send Through GSM → Enter OTP → Compare + Check RTC → Accept / Reject

7. Final Startup State

Initialization Result	System State
LCD ready	Display available for prompts/status messages.
Keypad ready	User input available.
EEPROM ready	User database and configuration can be accessed.
GSM ready	OTP SMS and authorized SMS command processing available.
RTC ready	OTP expiry timing available.

Initialization Result	System State
Door interface ready	L293D/DC motor control available when access is granted.
System ready	LOGIN prompt displayed; system waits for User ID.

Startup flow: POWER ON → LCD → Keypad → SPI/EEPROM → Management Interrupt → UART0 → GSM → RTC → OTP Module → User Database → SYSTEM READY → LOGIN

8. Initialization Checklist

- LPC2148 powers up and program starts
- LCD initializes and displays startup message
- Keypad scanning works
- SPI communication with EEPROM works
- EEPROM diagnostic passes
- Management switch / external interrupt is configured
- UART0 communicates at 9600 bps
- GSM modem responds to AT command
- SMS text mode is configured
- Incoming SMS mechanism is configured
- RTC time is valid
- OTP module is initialized
- User database loads correctly
- L293D/DC motor interface is ready
- LOGIN prompt is displayed

Project: SecureEntry – GSM-Based Multi-Level Secure Door Access System **Controller:** LPC2148 ARM7 **Language:** Embedded C